

(Section A)

Section A consists of 20 questions of 1 mark each.

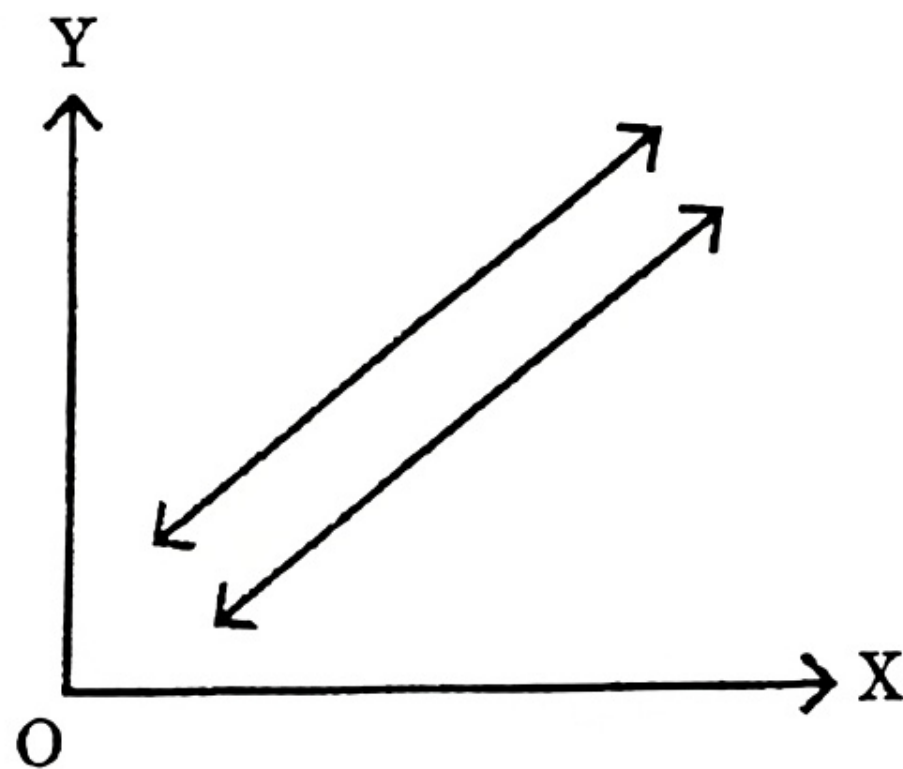
1. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$; x, y are prime numbers, then HCF (a, b) is [1]

A. xy
B. xy^2
C. x^3y^3
D. x^2y^2

2. The number of polynomials having zeroes as -2 and 5 is [1]

A. 1
B. 2
C. 3
D. more than 3

3. Given below is a graph showing two lines that never intersect. These are represented by two linear equations. [1]



Which of these can be said about the number of solution(s) of the above pair of linear equations?

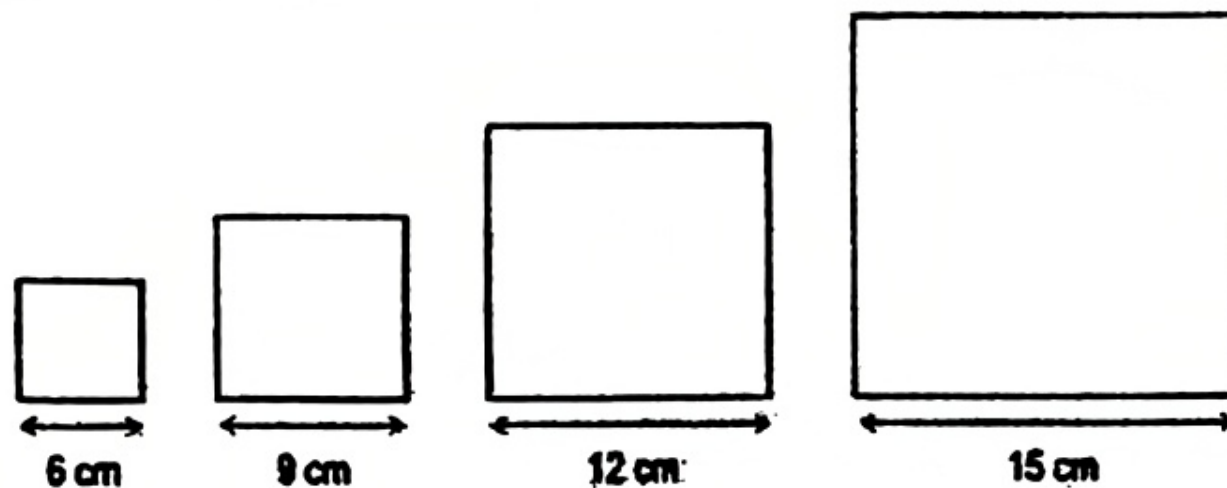
- A. They have infinitely many solutions.
B. They have a unique solution.
C. They do not have a solution.

D. Nothing can be said about the number of solutions unless the algebraic form of these equations are known.

1. The roots of quadratic equation $3x^2 - 4\sqrt{3}x + 4 = 0$ are [1]

- A. not real
- B. real and equal
- C. rational and distinct
- D. irrational and distinct

5. Shown below are some squares whose sides form an arithmetic progression (AP). [1]



Which of these are also in AP?

- i) The areas of these squares.
- ii) The perimeters of these squares.
- iii) The length of the diagonals of these squares.

- A. only ii)
- B. only i) and ii)
- C. only ii) and iii)
- D. all – i), ii) and iii)

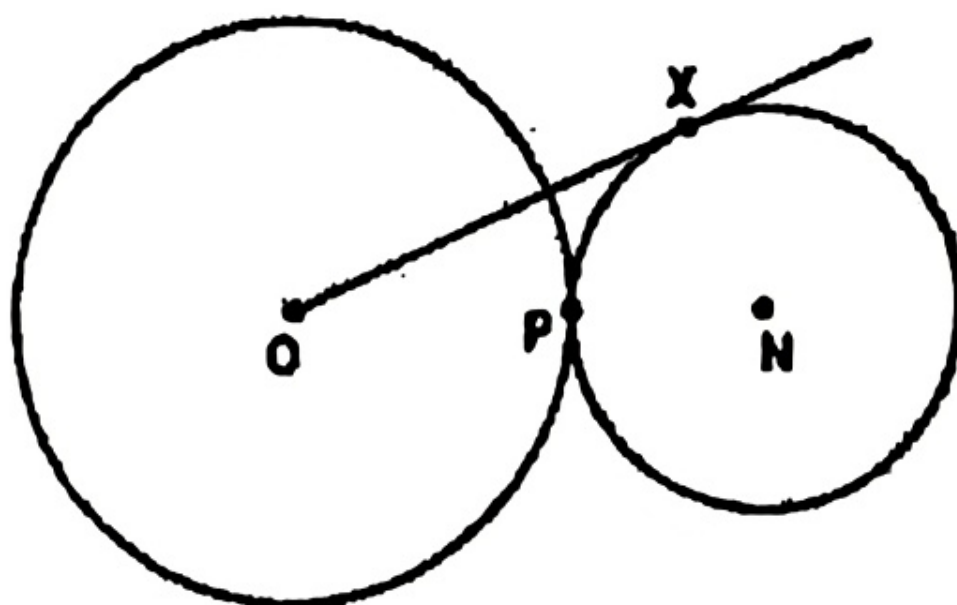
6. A line intersects the y-axis and x-axis at the points P and Q, respectively. If $(2, -5)$ is the mid-point of PQ, then the coordinates of P and Q are, respectively [1]

- A. $(0, -5)$ and $(2, 0)$
- B. $(0, 10)$ and $(-4, 0)$
- C. $(0, 4)$ and $(-10, 0)$
- D. $(0, -10)$ and $(4, 0)$

7. Which one of these is the relation between x and y if (x, y) is equidistant from $(-1, 4)$ and $(2, 5)$? [1]

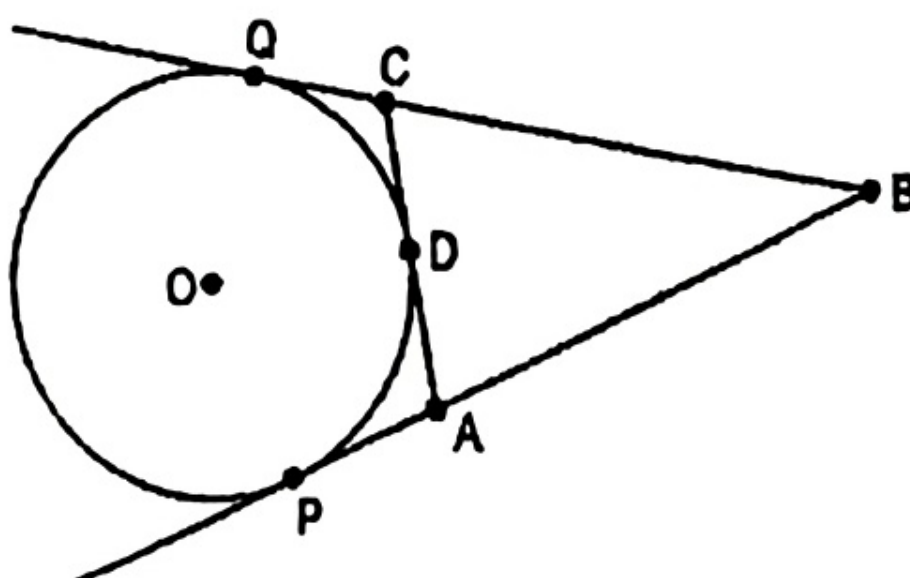
- A. $3x - y = 6$
 B. $6x + 2y = -9$
 C. $3x + y = 6$
 D. $3x - y = 3$

8. Two circles with centres O and N touch each other at point P as shown. O, P and N are collinear. The radius of the circle with centre O is twice that of the circle with centre N . OX is a tangent to the circle with centre N , and $OX = 18$ cm. What is the radius of the circle with centre N ? [1]



- A. $18/\sqrt{2}$ cm
 B. 9 cm
 C. $9/\sqrt{2}$ cm
 D. $18/\sqrt{10}$ cm

9. In the figure below, $\triangle ABC$ is formed using three tangents to a circle centred at O . [1]



(Note : The figure is not to scale.)

Based on the construction, which of the following statements is true?

- A. The sum of the length of BP and BQ less than the perimeter of $\triangle ABC$.
- B. The sum of the length of BP and BQ is same as the perimeter of $\triangle ABC$.
- C. The sum of the length of BP and BQ is greater than the perimeter of $\triangle ABC$.
- D. Length of the tangents should be known to compare it to the perimeter of $\triangle ABC$.

10. If $\triangle ABC$ is right angled at C, then the value of $\cos (A+B)$ is [1]

- A. 0
- B. 1
- C. $\frac{1}{2}$
- D. $\frac{\sqrt{3}}{2}$

11. $\sec^4 \theta - \sec^2 \theta =$ [1]

- A. $\tan^4 \theta - \tan^2 \theta$
- B. $\tan^2 \theta - \tan^4 \theta$
- C. $\tan^4 \theta + \tan^2 \theta$

~~D. none of these~~ $\tan^3 \theta + \tan^2 \theta$

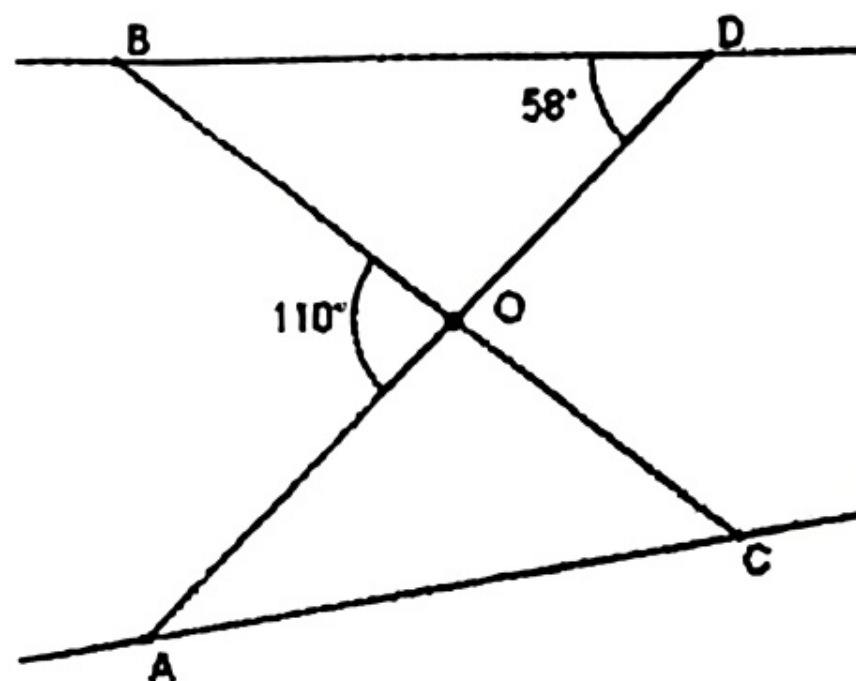
12. The top of two poles of height 10 m and 16 m are connected by a wire. If the wire makes an angle of 30° with the horizontal, then the length of the wire is [1]

- | | |
|---------|---------|
| A. 12 m | B. 16 m |
| C. 10 m | D. 26 m |

13. If the circumference of a circle and the perimeter of a square are equal, then [1] 16

A. Area of the circle = Area of the square
 B. Area of the circle > Area of the square
 C. Area of the circle < Area of the square
 D. Nothing definite can be said about the relation between the areas of the circle and square.

14. In the given figure, if $\triangle OCA \sim \triangle OBD$ then $\angle OAC$ is equal to [1] 1



A. 58°
 B. 55°
 C. 128°
 D. 52°

15. Which of the following information is NOT sufficient to differentiate between minor and major sector of a circle? [1]

A. the angle subtended at the centre
 B. the radius of the circle
 C. the area of the sector
 D. the length of the arc

16. There are 10 observations in ascending arrangements. If last 5 observations are increased by 8, then Median [1] 19

A. Increase by 8
 B. Increase by 4
 C. Remains same
 D. Both a and b

- A girl calculates that the probability of her winning the first prize in a lottery is 0.08. If 6000 tickets are sold, how many tickets has she bought? [1]

A. 40
B. 240
C. 480
D. 750

- 18: Pratik has blue and green coins of the same size in a bag. He has 50 coins each of blue and green. He is randomly picking up one coin at a time without replacement. He does not see which coin he has picked. What is the MINIMUM number of coins he would have to pick to definitely get a pair of blue or green coins? [1]

A. 2
B. 3
C. 4
D. 5

Direction for questions 19 & 20 : In question numbers 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

- A. Both Assertion (A) and Reason(R) are true and Reason(R) is the correct explanation of Assertion (A).
B. Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
C. Assertion (A) is true but Reason(R) is false.
D. Assertion (A) is false but Reason(R) is true.

19. Assertion (A) : "The sequence $-1, -1 - 1, \dots, -1$ is an AP."

Reason (R) : "In an AP, $a_n - a_{n-1}$ is constant where $n \geq 2$ and $n \in N$ " [1]

20. Assertion (A) : "The volume of the largest right circular cone that can be cut out of a cube whose edge is 7 cm is 50 cm^3 ."

Reason (R) : "Volume of a cone with radius r and height is $\frac{1}{3}\pi r^2 h$."

[1]

(Section – B)

Section B consists of 5 questions of 2 marks each.

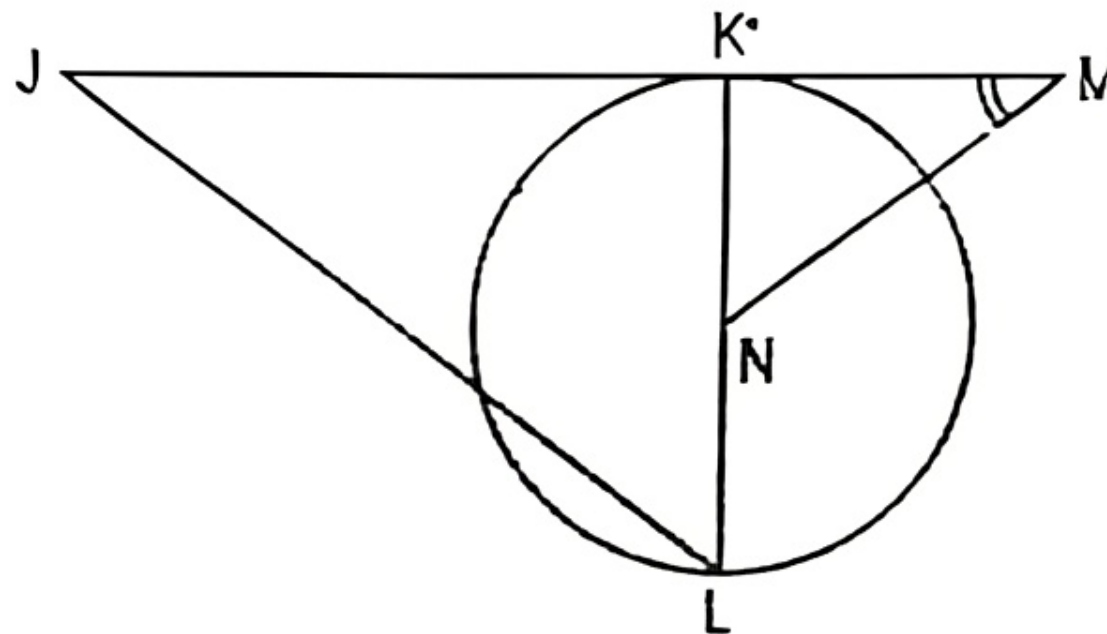
21. $\sqrt{5}$ is an irrational number. Meera was asked to prove that $(3 + \sqrt{5})$ is an irrational number. Shown below are the steps of Meera's proof :

[2]

Step 1	Let $(3 + \sqrt{5})$ be a rational number. Then $(3 + \sqrt{5})$ can be written as $\frac{p}{q}$ where p and q ($q \neq 0$) are co-primes.
Step 2	Hence, $\sqrt{5} = \left(\frac{p}{q} - 3\right)$.
Step 3	Since p and q are integers, $\left(\frac{p}{q} - 3\right)$ is also an integer.
Step 4	Since $\left(\frac{p}{q} - 3\right)$ is an integer and every integer is a rational number, $\left(\frac{p}{q} - 3\right)$ is a rational number. It implies that $\sqrt{5}$ is a rational number.
Step 5	But this contradicts the fact that $\sqrt{5}$ is an irrational number. Hence, $(3 + \sqrt{5})$ is an irrational number.

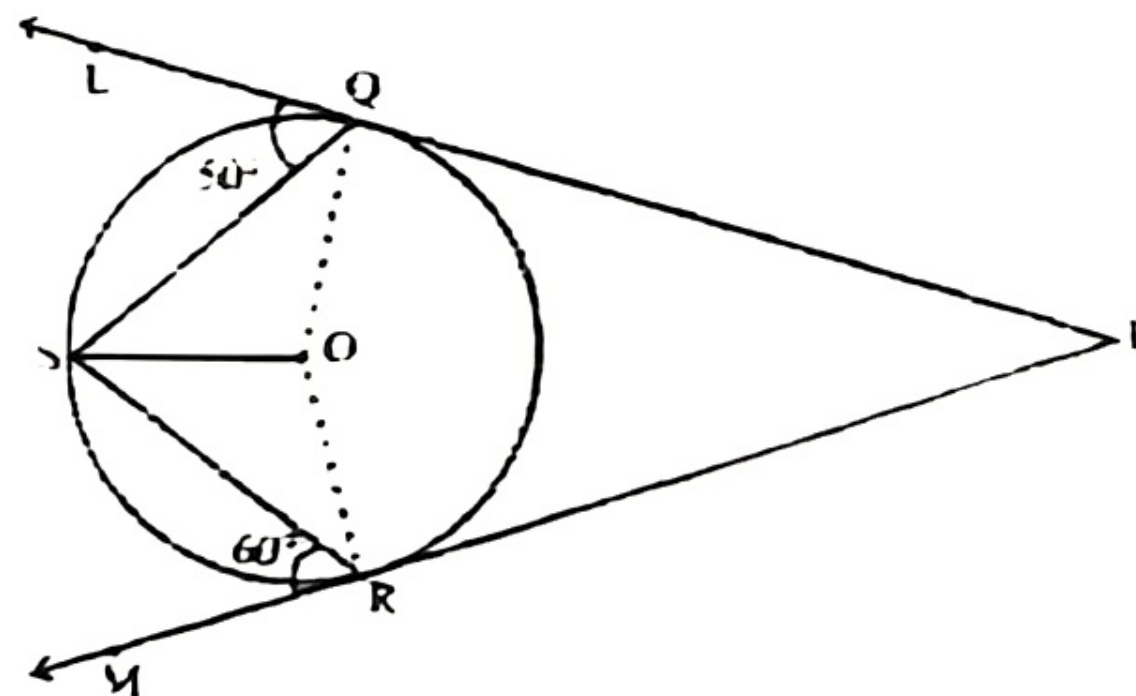
She made an error in one step due to which her subsequent steps were incorrect too. In which step did she make that error? Justify your answer.

22. In the figure below, JM is tangent to the circle which has its centre at point N and $\angle LJK = \angle NMK$. [2]



If $JL = 15$ cm, find the length of MN. Show your work.

23. In the given figure, PQL and PRM are tangents to the circle with centre O at the points Q and R, respectively and S is a point on the circle such that $\angle SQL = 50^\circ$ and $\angle SRM = 60^\circ$. Then show that $\angle QSR$ is equal to ~~40°~~ 70°. [2]

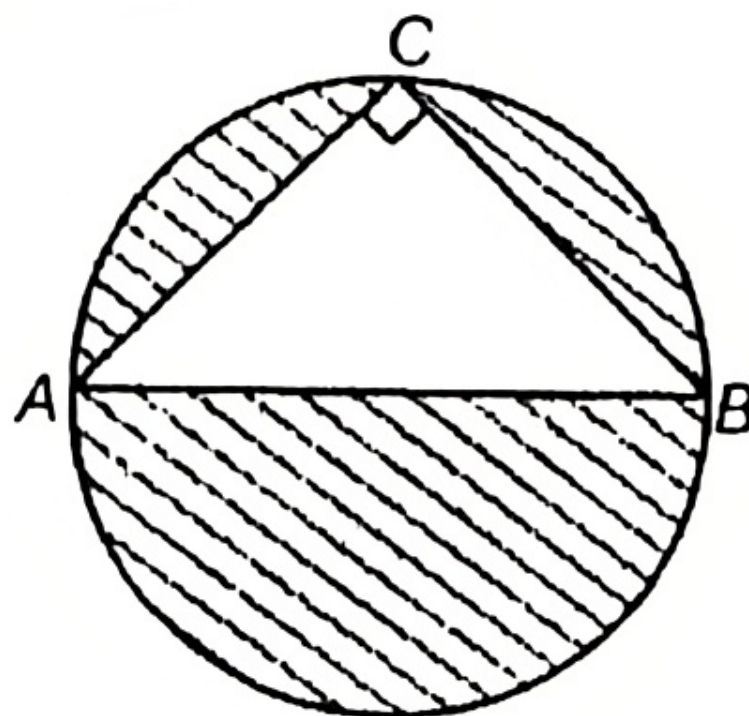


24. A. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, $0^\circ < A + B < 90^\circ$ and $A > B$, then find the values of A and B. [2]

OR

B.⁰ Find the value of the expression
 $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$

25. A.['] In the given figure, AB is a diameter of the circle, AC = 6 cm and BC = 8 cm. Find the area of the shaded region (Use $\pi = 3.14$). [2]



OR

B. The wheel of a motor cycle is of radius 35 cm. How many revolutions per minute must the wheel make so as to keep a speed of 66 km/h?

(Section – C)

Section C consists of 6 questions of 3 marks each

26. ¹⁰ On a morning walk, three persons step off together and their steps measure 40 cm, 42 cm and 45 cm, respectively. What is the minimum distance each should walk so that each can cover the same distance in complete steps? [3]

27. Aasira multiplied a variable with 4, subtracted 12 and added the square of the original variable. She expressed the final expression as a product of 2 factors. Her friend, Rishi, said that the factors will always have a difference of 8. Is Rishi right? Show your work. [3]

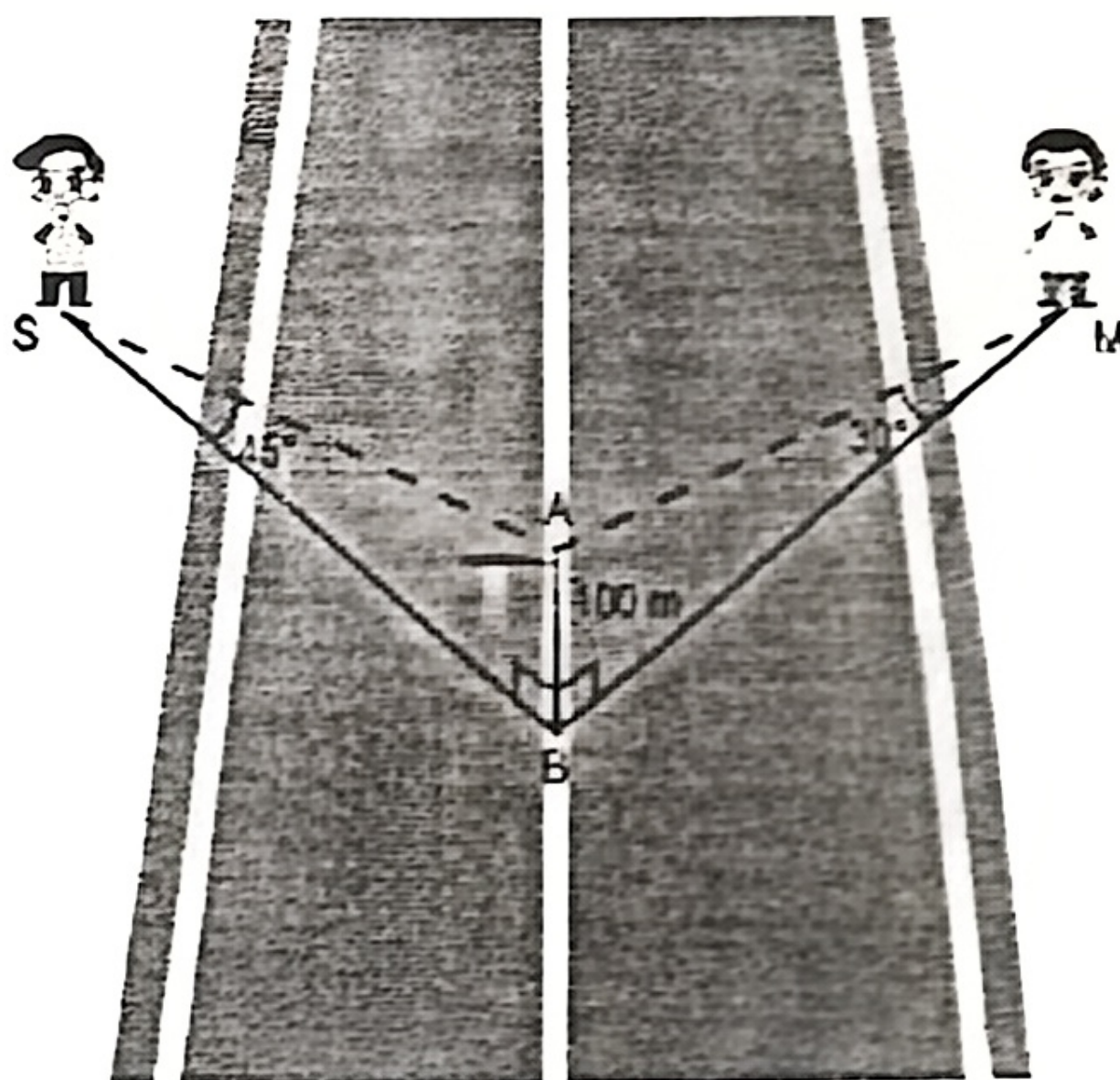
28. A. Draw the graph of the pair of equations $2x + y = 4$ and $2x - y = 4$. Write the vertices of the triangle formed by these lines and the y-axis. [3]

OR

- B. Two numbers are in the ratio $5 : 6$. If 8 is subtracted from each of the numbers, the ratio becomes $4 : 5$. Find the numbers.
29. A. Prove that the parallelogram circumscribing a circle is a rhombus. [3]

OR

- B. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
30. Shoaib and Mihir are playing in their school playground which has a 100 m light pole. Shoaib is standing towards the north of the light pole and Mihir is standing towards the east of the light pole. From the point where Shoaib and Mihir stand, the angle of elevation of the top of the pole is 45° and 30° respectively, as shown in the image below.



Find the shortest distance between Shoaib and Mihir. Show your work.

31. Recast the following cumulative "More than" table in the form of an ordinary frequency distribution and determine the median [2]

Number of days absent	Number of Students
More than 0	120
More than 5	90
More than 10	60
More than 15	30
More than 20	0

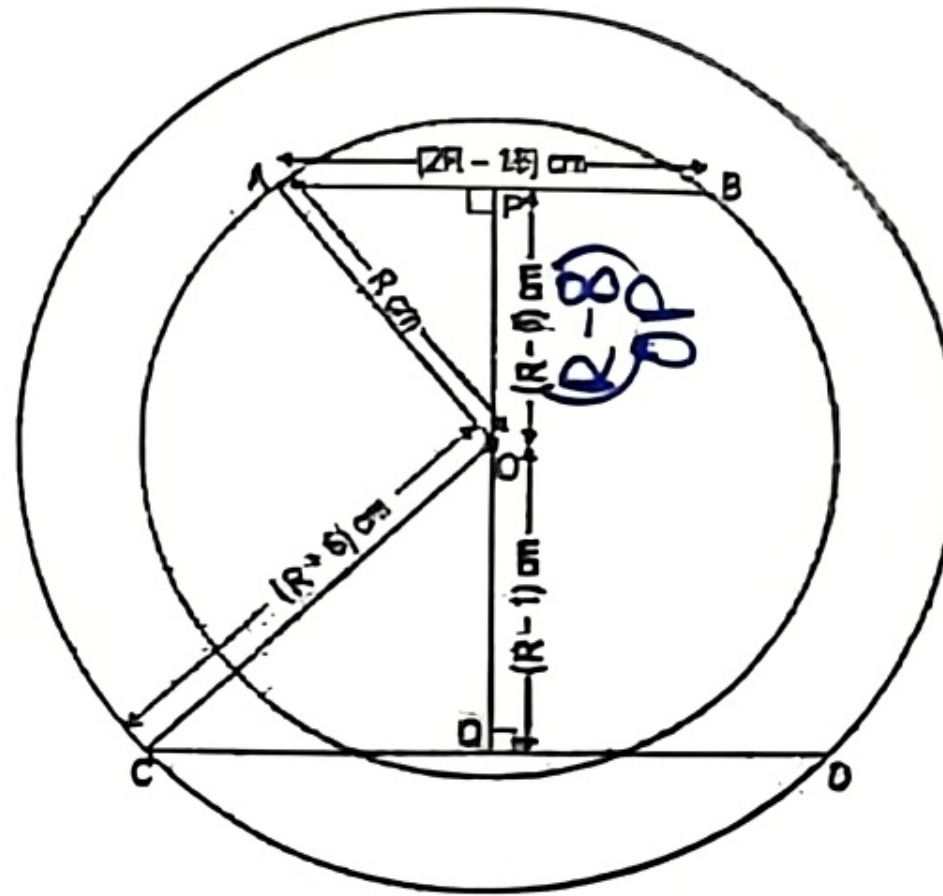
(Section - D)

Section D consists of 4 questions of 5 marks each

32. A. Vitthal went on a long cycling round trip from his home to a nearby hill and back. The journey to the hill started with a 40 km flat road and then a 25 km uphill slope. While going uphill, his average speed reduced by 10 km/hr compared to the flat road leading to an extra half-hour of travel time. However, he maintained the same average speed on the flat road in both directions.
- Find the time taken by Vitthal to reach the end of 65 km stretch in the uphill direction.
 - If he took $2\frac{1}{2}$ hrs to reach home from the hill, find his average speed on the slope while descending.

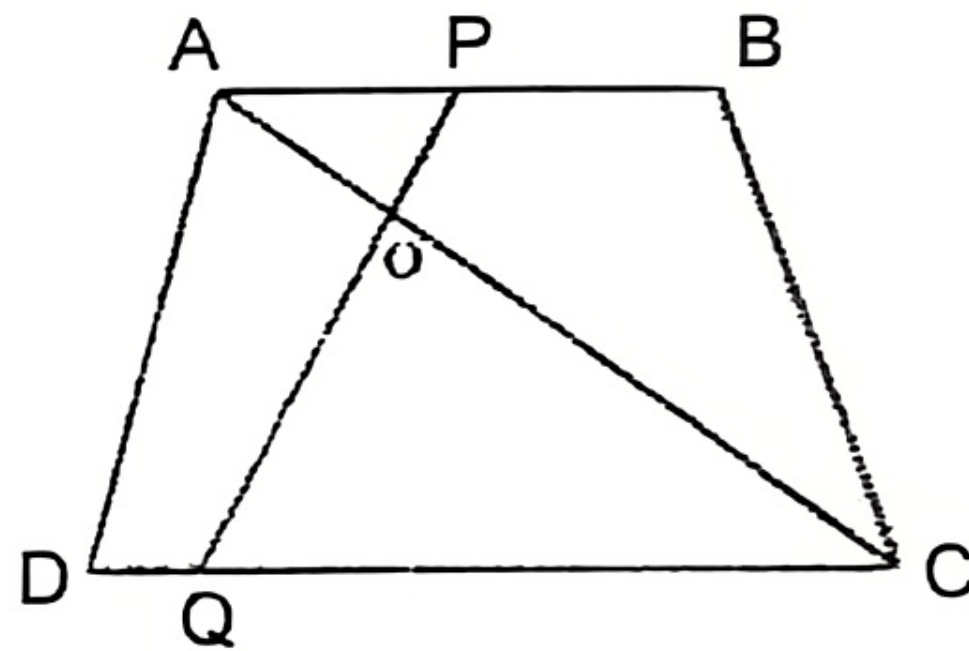
OR

- B. In the figure below, two concentric circles have centre O. Their radii are R and (R + 6) cm respectively.



Find the lengths of the chords AB and CD. Show your work.

33. In the given figure, if $AB \parallel DC$ and AC and PQ intersect each other at the point O, prove that $OA.CQ = OC.AP$.



34. A. A restaurant stores ice-cream in a box with a dispenser attached for filling ice-cream cones. The dimensions of the box and the ice-cream cones used by the restaurant are shown in Figure 1 below. To make each serving of dessert, the cone is first filled with ice-cream and then topped with a hemispherical scoop of ice-cream taken from the same box, as shown in Figure 2. [5]

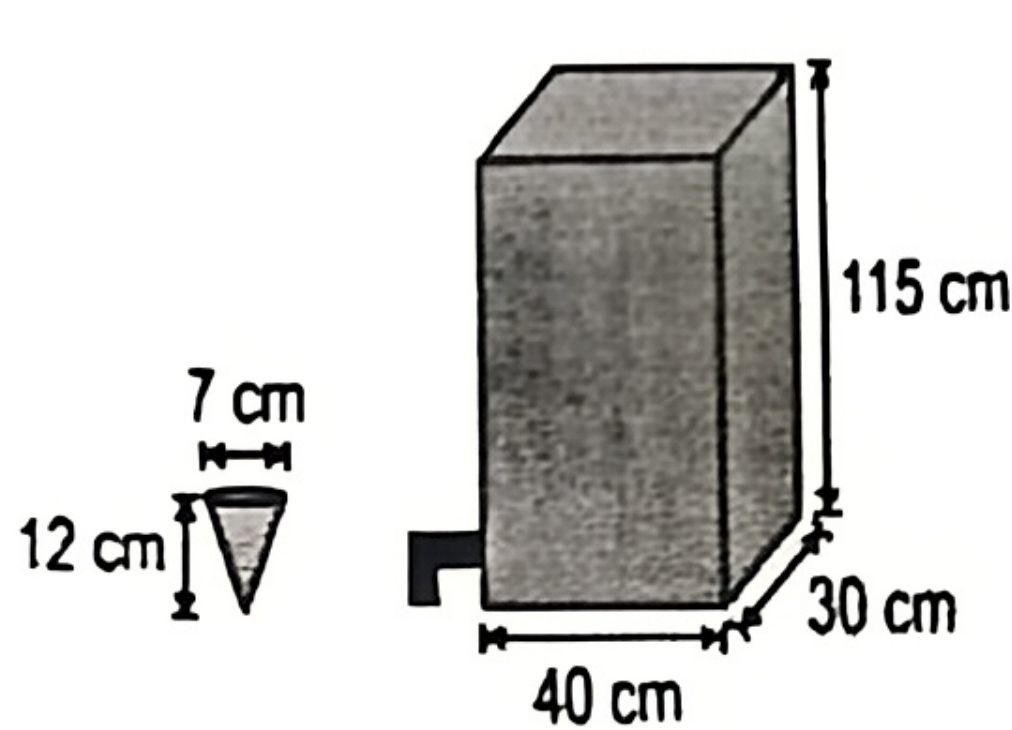


Figure 1



Figure 2

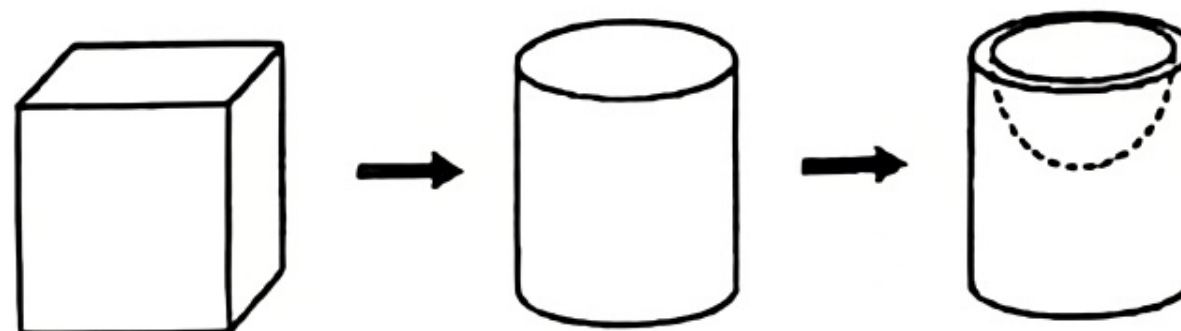
35

Approximately how many desserts can be served out of a completely filled box of ice-cream? Show your steps. (Note : Take π as $22/7$)

OR

- B. Aparna takes a wood carving class during her summer camp. She attempts to make a bird feeder. She takes a piece of wood in the shape of a cube and carves it to make a cylinder with height and diameter equal to the side of the cube. She then carves a hemisphere into the circular base of the cylinder with radius equal to $\frac{4}{5}$ th of the radius of the cylinder. After this, she sands the wood to make it smooth.

36



(Note : The figure is not to scale.) What percentage of the original cube has been used to make the bird feeder? Show your work.

(Note : Take π as 3.14. Round your answer to the nearest integer.)

35. The monthly expenditure on milk in 200 families of a Housing Society is given below [5]

Monthly Expenditure (in Rs.)	1000-1500	1500-2000	2000-2500	2500-3000	3000-3500	3500-4000	4000-4500	4500-5000
Number of families	24	40	33	X	30	22	16	7

Find the value of x and also find the mean expenditure

28

2662.5

(Section – E)

Section E consists of 3 case study-based questions of 4 marks each.

36. A thief, after committing a theft, runs on his bike at a speed of 100 m/min and increases his speed by 2 m/min every succeeding minute. After one minute, a policeman runs on his jeep to catch him. He goes 100 m in the first min and increases his speed by 20 m/min every succeeding minute.



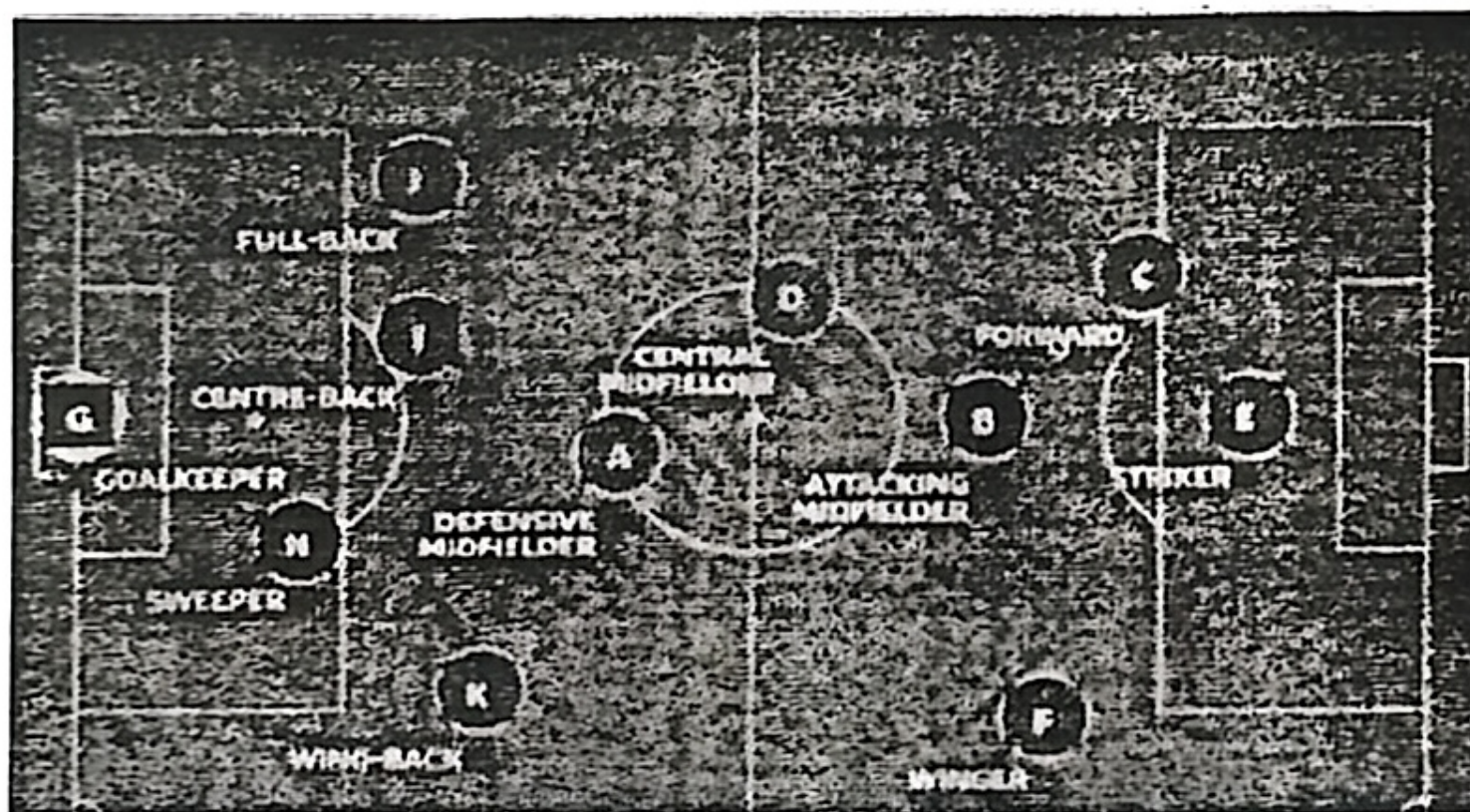
Based on the above information, answer the following questions :

- (i) Find the distance travel by thief in n^{th} minute? [1]
- (ii) Find the total distance travel by policeman after p^{th} minute? [1]
- (iii) A.* After how many minutes policeman will catch the thief? [2]

OR

- B. If policeman runs on his jeep with uniform speed of 121 m/min. Then after how many minutes he will catch the thief?

37. Tharunya was thrilled to know that the football tournament is fixed with a monthly timeframe from 20th July to 20th August 2023 and for the first time in the FIFA Women's World Cup's history, two nations host in 10 venues. Her father felt that the game can be better understood if the position of players is represented as points on a coordinate plane.



Based on the above information, answer the following questions :

- (i) At an instance, the midfielders and the forward formed a parallelogram. Find the position of the central midfielder (D) if the position of other players who formed the parallelogram are: A(1, 2), B(4, 3) and C(6, 6). [1]

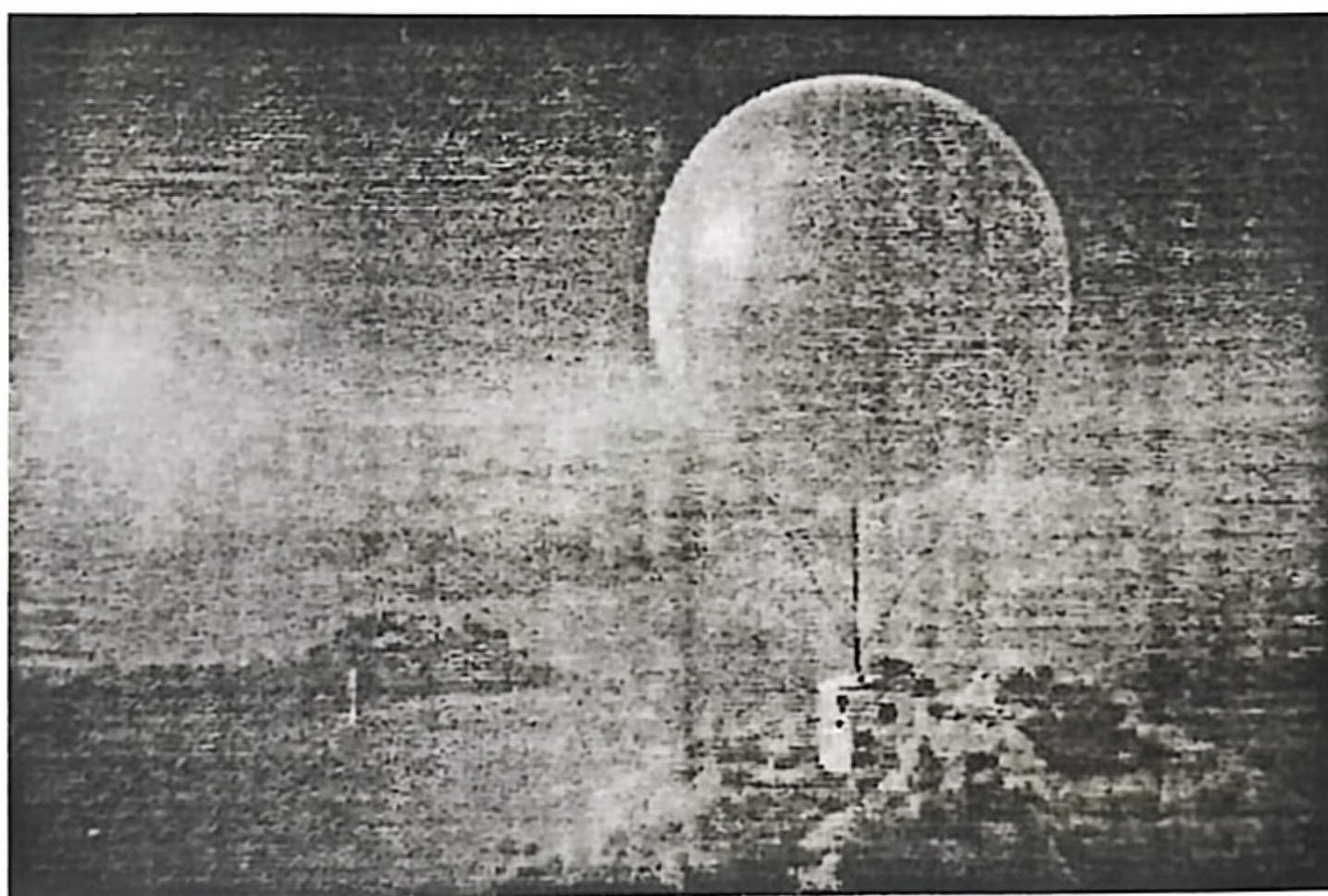
- (ii) A. Check if the Goal keeper $G(-3, 5)$, Sweeper $H(3, 1)$ and Wing-back $K(0, 3)$ fall on a same straight line. [2]

OR

- B. Check if the Full-back $J(5, -3)$ and Centre-back $I(-4, 6)$ are equidistant from forward $C(0, 1)$ and if C is the mid-point of JI .

- (iii) If Defensive midfielder $A(1, 4)$, Attacking midfielder $B(2, -3)$ and Striker $E(a, b)$ lie on the same straight line and B is equidistant from A and E , find the position of E . [1]

38. A transparent spherical balloon of radius r subtends an angle α at the eye of an observer. If the angle of elevation of its centre is β .



- (i) Draw an appropriate figure for above situation? [1]
- (ii) Distance of centre of spherical balloon from the eye of observer? [1]
- (iii) A. What is height of centre of spherical balloon from the ground? [2]

[18]

OR

B.

A transparent spherical balloon of radius 1.5m subtends an angle 60° at the eye of an observer of height 3m. If the angle of elevation of its centre is 45° . What is height of centre of spherical balloon from the ground?